ADRENALINE

High-alert drug

Check concentration & route

CHEMICAL NAME (R)-4-[1-hydroxy-2-(methylamino)ethyl]benzene-1,2-diol (as tartrate or hydrochloride); also called epinephrine

CLASS Sympathomimetic catecholamine (α and β agonist)

TRADE NAMES EpiPen / Jext (autoinjector) · Adrenalin · Adrenaline 1:1000 and 1:10,000

THERAPEUTIC USES / INDICATIONS

- Anaphylaxis (first-line, IM).
- Cardiac arrest (all rhythms).
- Severe hypotension/septic shock (infusion, often second-line).
- Severe croup and bronchiolitis (nebulised); with local anaesthetics (vasoconstriction); topical haemostasis.

PHARMACODYNAMICS

- **α1:** vasoconstriction → ↑ BP, ↓ mucosal oedema.
- **β1:** ↑ heart rate and contractility.
- **β2:** bronchodilation; stabilises mast cells (↓ histamine release).

ADMINISTRATION

- **Anaphylaxis:** IM into anterolateral thigh using 1 mg/mL (1:1000); repeat after 5 min if needed.
- **Cardiac arrest:** IV/IO 1 mg of 1:10,000 (0.1 mg/mL) every 3–5 min.
- **Infusion:** via central line with syringe pump (extravasation → necrosis).
- Never give 1:1000 IV as bolus — fatal arrhythmias.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tachycardia, palpitations, tremor.
- Anxiety, pallor, headache.
- Hyperglycaemia, hypokalaemia.

ADVERSE EFFECTS (SERIOUS)

- Ventricular arrhythmias; myocardial ischaemia.
- Severe hypertension, cerebral haemorrhage (overdose/IV error).
- Tissue necrosis with extravasation.
- Lactic acidosis (infusion).

DOSAGES

- **Anaphylaxis IM (1 mg/mL):** adult/child >12 yr 0.5 mg (0.5 mL); 6–12 yr 0.3 mg; 6 months–6 yr 0.15 mg (US label: <30 kg 0.01 mg/kg).
- **Cardiac arrest:** adult 1 mg IV/IO every 3–5 min; child 10 mcg/kg (0.1 mL/kg of 1:10,000).
- **Infusion (septic shock):** 0.05–2 mcg/kg/min titrated to BP; wean gradually.
- **Nebulised (severe croup, off-label):** per paediatric protocol.

CONTRAINDICATIONS & CAUTIONS

- None in anaphylaxis or cardiac arrest (benefit outweighs risk).
- **Caution:** coronary disease, hypertension, hyperthyroidism, diabetes, angle-closure glaucoma; avoid in end-artery areas (local use).

KEY INTERACTIONS

- **Non-selective β-blockers:** severe hypertension, reflex bradycardia; reduced response in anaphylaxis (glucagon may help).
- **TCAs, MAOIs:** ↑ pressor effect.
- **Volatile anaesthetics:** arrhythmias.

PHARMACOKINETICS

- **Onset:** IV immediate; IM 5–10 min. **Duration:** IV 2–5 min; IM 20–40 min.
- Rapidly metabolised by COMT and MAO.
- Metabolites excreted in urine.

NURSING MANAGEMENT

- Confirm concentration (1:1000 vs 1:10,000) and route before giving.
- Anaphylaxis: lie patient flat (legs raised), give IM promptly, call for help, give oxygen, IV fluids.
- Monitor ECG, BP, HR, SpO_2; record time of each dose.
- Infusion: central line, check site, titrate to MAP.
- Observe after anaphylaxis for biphasic reaction (6–12 h).

PATIENT EDUCATION

- Carry two autoinjectors at all times; check expiry dates.
- Use in outer thigh (through clothing), hold 3–10 seconds, then call an ambulance.
- Lie down with legs raised (sit if breathing difficult).
- Train family/school staff in its use.

KEY POINTS

- α + β agonist — first-line for anaphylaxis (IM 0.5 mg).
- Check concentration; 1:1000 never IV bolus.
- Arrest: 1 mg IV every 3–5 min.
- Biphasic reactions — observe after anaphylaxis.

SOURCES CHECKED **[376]** Adrenaline (Epinephrine) 1mg/ml (1:1000) solution for injection (ampoule) (UK SmPC)
[377] Adrenaline 1 mg/10 ml (1:10,000), solution for injection in pre-filled syringe (UK SmPC) **[378]** Epinephrine, IM, IV, SC (US label)

Full references p. 432–457